# Advanced SQL

## Exercise 5

## NULL functions

### Q1

Show all employees with their Salary. If Salary is Null, display 0.

```
SELECT    employee_id,
          name
IFNULL (Salary, 0) AS Salary_with_default
FROM      Employees;
```

Outcome

| employee_id | name | Salary_with_default |
|---|---|---|
| 1 | Alice | 5000 |
| 2 | Bob | 0 |
| 3 | Charlie | 7000 |
| 4 | Dana | 0 |

Q2

Show employee names with their department. If department is NULL, Show 'Not Assigned'

```
SELECT employee_id,
       name
IFNULL (department, 'Not Assigned') AS department_name
FROM Employees ;
```

Outcome

| employee_id | name | Department_name |
|---|---|---|
| 1 | Alice | HR |
| 2 | Bob | IT |
| 3 | Charlie | Not Assigned |
| 4 | Dana | Finance |

Q3

Find orders with NULL customer_id using IS NULL ().

```
SELECT  order_id,
        customer_id
FROM    orders
WHERE   (customer_id) IS NULL;
```

Outcome

| Order_Id | Customer_Id |
|---|---|
| 103 | NULL |

Q4

Show all orders. If delievery_date is NULL, Show 'Pending'.

```
SELECT      order_Id,
            Customer_Id
COALESCE (delivery_date, 'Pending') AS delievery_Status
FROM        orders;
```

Outcome

| Order_Id | Customer_Id | delivery_Status |
|---|---|---|
| 101 | 201 | 2024-12-01 |
| 102 | 202 | Pending |
| 103 | NULL | 2024-12-03 |

## Q5

Show all Students and their grades. Replace NULL with 0.

```
SELECT  Student_Id,
        name
IFNULL  (grade, 0) AS final_grade
FROM    Students;
```

Outcome

| Student_Id | name | final_grade |
|---|---|---|
| 1 | Ethen | 85 |
| 2 | Maya | 0 |
| 3 | Olivia | 90 |

Q6

Count Students who haven't been graded.

```
SELECT  COUNT(*) AS not_graded_Count
FROM    Students
WHERE   grade IS NULL;
```

Outcome

| not_graded_Count |
| --- |
| 1 |

Q7

Show product name, price and final price after discount (assume 0 if discount is NULL).

```
SELECT product_id,
       name,
       price
IFNULL (discount, 0) AS final_price
FROM products;
```

| Product_id | name | final_price |
|---|---|---|
| 501 | Keyboard | 25 |
| 502 | Mouse | 10 |
| 503 | Monitor | 100 |

Q8. Co

Count how many customers have no email.

```
SELECT  COUNT(*) AS missing_email_Count
FROM    Customers
WHERE   email IS NULL;
```

Outcome

| Missing_email_Count |
| --- |
| 2 |

Q9.

Show all customers with email, if Null, display 'No email'.

```
SELECT    customer_id,
          name
COALESCE (email, 'No email') AS display_email
FROM      customers;
```

Outcome

| customer_id | name | email_display |
|---|---|---|
| 1 | Linda | No email |
| 2 | Joseph | Joseph@mail.com |
| 3 | Nia | No email |

Q10 :-

Show payment details with method replaced by 'unknown' if NULL.

```
SELECT  payment_id, Status
IFNULL  (method, 'unknown')  method_display
FROM    Payments ;
```

| ~~Outcome~~ Payment_id | ~~pa~~ method_display | Status |
|---|---|---|
| 301 | Credit | NULL |
| 302 | Paypal | Success |
| 303 | Unknown | Failed. |

Q11 →

Show items and their quantity (or NULL).

```
SELECT    item_id,
          item_name
IFNULL    (quantity, 0) AS quantity_check
FROM      inventory;
```

Outcome

| Item_id | item_name | quanity_check |
|---|---|---|
| 1 | Pen | 0 |
| 2 | Notebook | 150 |
| 3 | Eraser | 0 |

Q12

Show employee_id and the first available value among bounous or Commission.

```
SELECT employee_id
COALESCE (bounus, Commission) As first_availible_reward
FROM Employee_Extra;
```

Outcome

| employee_id | first_available_reward |
|---|---|
| 1 | 300 |
| 2 | 100 |
| 3 | NULL |

Q13

Count classes that don't have a room assigned

```
SELECT   COUNT(*) AS no_room_assigned
FROM     classes
WHERE    room IS NULL;
```

Outcome

| no_room_assigned |
| --- |
| 2 |

Q14 →

Show attendance records with Status. Replace NULL with 'Not Marked'.

```
SELECT  Student_Id,
        date
IFNULL  (Status, 'Not Marked') AS attendance_Status
FROM    Attendance;
```

Outcome

| Student_Id | date | attendance_status |
|---|---|---|
| 1 | 2025-04-01 | Not Marked |
| 2 | 2025-04-01 | Present |
| 3 | 2025-04-01 | Absent |

Q15

Show accounts_id, account_type (or 'Unknown'), and balance (or 0).

```
SELECT account_id
IFNULL (account_type, 'Unknown') AS type_display
IFNULL (balance, 0) AS balance_checked
FROM Bank_accounts;
```

Outcome

| account_id | type_display | balance_checked |
|---|---|---|
| A1 | Savings | 0 |
| A2 | Current | 5000 |
| A3 | Unknown | 2000 |

Q16

Show all projects with a start date, if start date is NULL, display 'TBD'.

```
SELECT  project_id,
        title,
IFNULL  (start_date, 'TBD') AS start_display
FROM    Projects ;
```

Outcome

| Project_id | title | start_display |
|---|---|---|
| 1 | website revamp | 2025-01-10 |
| 2 | Mobile App | TBD |
| 3 | Data migration | TBD |

Q17

Display reviews showing comments (or 'no comment') and rating (or 0).

```
SELECT  review_id,
        Product_id
IFNULL (comment, 'no comment') AS comment_display,
IFNULL (rating, 0) AS rating_display
FROM Reviews;
```

Outcome

| review_id | product_id | comment_display | rating_display |
|---|---|---|---|
| 1 | 501 | Great product | 4 |
| 2 | 502 | No comment | 0 |
| 3 | 503 | works fine | 3 |

Q15.

Show the Supplier Contact number. Use COALESCE (phone, alt_phone, 'no contact').

```
SELECT    Supplier_id,
          name
COALESCE (phone, alt_phone, 'no contact') AS Contact_number
FROM      Suppliers;
```

Outcome

| Supplier_id | name | Contact_number |
|---|---|---|
| 1 | Global Goods | 123456789 |
| 2 | Best Supplier | 9876543321 |
| 3 | Value Source | No contact. |

Q19

Show all users and their preferences. Replace NULLS with defaults: • Theme → "Light" • Language → "English" • Timezone → "UTC"

```
SELECT user_id
COALESCE (theme, 'Light') AS theme_set,
COALESCE (language, 'English') AS language_set,
~~FROM~~ COALESCE (timezone, 'UTC') AS timezone_set,
FROM user_settings;
```

Outcome

| user_id | theme_set | language_set | timezone_set |
|---|---|---|---|
| 1 | Light | English | UTC |
| 2 | Dark | English | UTC +1 |
| 3 | Light | English | UTC |

Q20

Show maintainance log with: • issue → default to 'unknown issue'
• technician → default to 'not Assigned'

```
SELECT    record_id,
          machine_id,
COALESCE (issue, 'unknown') AS issue_log
COALESCE (technician, 'not Assigned') AS technician_name
FROM Maintainance
```

Outcome

| record_id | machine_id | issue_id | technician_name |
|---|---|---|---|
| 1 | M101 | Overheating | Not Assigned |
| 2 | M102 | Unknown issue | Not Assigned |
| 3 | M103 | ~~Jamming~~ Jammed | Alex |